Versorgung

Versorgung: 1S-Pfad (Elektronik) und 2S-Pfad (Aktoren, Pyro)

Datei: power.kicad_sch

Teensy

Teensy 4.1 auf Buchsenleisten, Erweiterungsleiste, Befestigung

Datei: mcu.kicad_sch

Sensorik

IMU ICM-42688-P an SPI1, Barometer BMP390 und Qwiic an I2C

Datei: sensors.kicad_sch

Funk

LoRa E22-400T22S an Serial8, GPS an Serial1

Datei: radio.kicad_sch

Aktoren

TVC-Servos am 2S-Akku, Fallschirm-Servo, Rollmotor DRV8833

Datei: actuators.kicad_sch

Pyro

Zwei Pyro-Kanaele low-side, Arming, Durchgangspruefung

Datei: pyro.kicad_sch

Signal

WS2812B mit Pegelwandler, Summer, Kamera-Trigger

Datei: signal.kicad_sch

## Kartonaut – Flugcomputer-Mainboard rev B (54 x 120 mm, 4 Lagen, JLCPCB)

Spezifikation: hardware/mainboard/README.md; Design Guide: docs/DesignGuide_Mainboard_revB.md

Massen: GND = Logik und 1S-Minus; GND_PWR = Rueckstrom aller 2S-Verbraucher. Einzige Verbindung: NT1 am 2S-Minus.

Dieser Schaltplan wird von gen_schematic.py aus circuit.py erzeugt. Aenderungen dort machen, dann neu erzeugen.

Teensy 4.1, ICM-42688-P, BMP390, E22-400T22S, 1S + 2S

**Kartonaut**

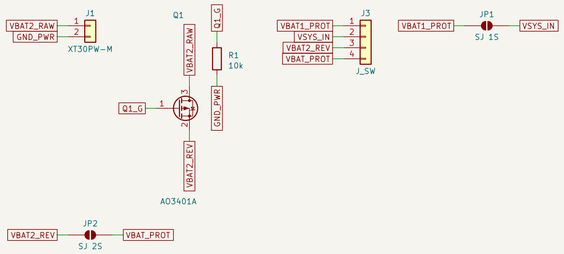
Sheet: /
File: mainboard.kicad_sch

***Title: Mainboard rev B***

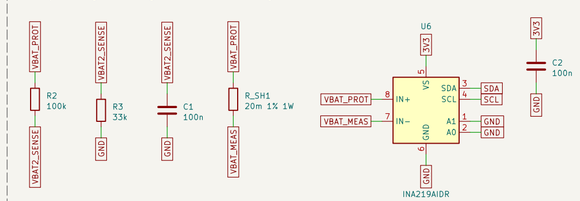
| Size: A3 | Date: 2026-10-21 | **Rev: B** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 1/8 |

# Versorgung: 1S–Pfad (Elektronik) und 2S–Pfad (Aktoren, Pyro)

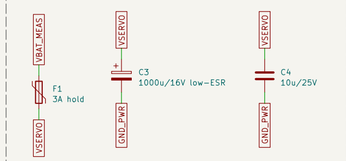
## 2S: Akku, Verpolschutz, Hauptschalter



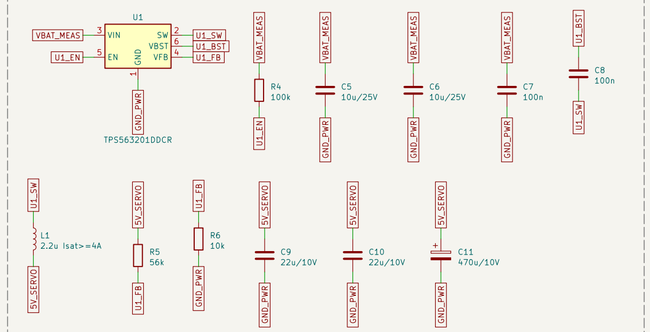

## 2S: Spannungsmessung und Strommessung INA219




## 2S: Servo–Schiene VSERVO




## 2S: Buck U1 –> 5V_SERVO




## 1S: Akku, Verpolschutz, Spannungsmessung



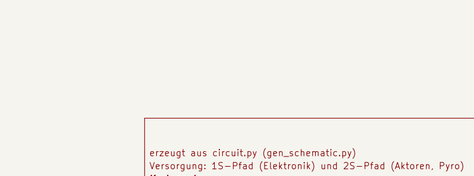

## 1S: Buck–Boost U2 –> 5V_SYS, Teensy–VIN




## LDO U3 –> 3V3_RF, Sternpunkt, Testpunkte




## Power–Flags fuer den ERC




erzeugt aus circuit.py (gen_schematic.py)
Versorgung: 1S–Pfad (Elektronik) und 2S–Pfad (Aktoren, Pyro)
**Kartonaut**

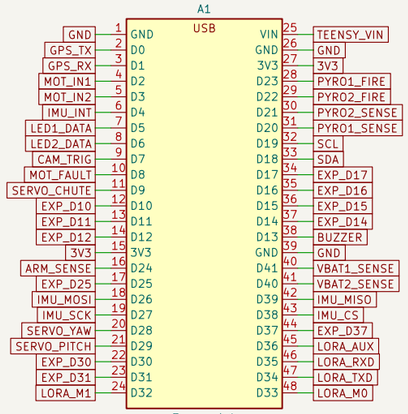
Sheet: /Versorgung/
File: power.kicad_sch

***Title: Mainboard rev B – Versorgung***

| Size: A2 | Date: 2026-10-21 | **Rev: B** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 2/8 |

# Teensy 4.1 auf Buchsenleisten, Erweiterungsleiste, Befestigung

## Teensy 4.1




## Erweiterung, Befestigung, Testpunkte




erzeugt aus circuit.py (gen_schematic.py)
Teensy 4.1 auf Buchsenleisten, Erweiterungsleiste, Befestigung
**Kartonaut**

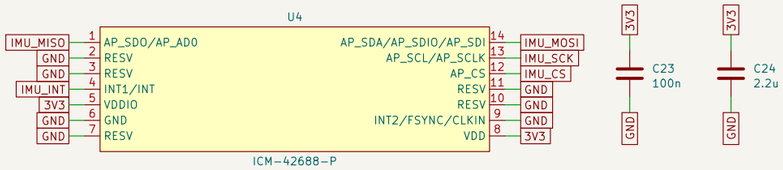
Sheet: /Teensy/
File: mcu.kicad_sch

***Title: Mainboard rev B – Teensy***

| Size: A3 | Date: 2026-10-21 | **Rev: B** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 3/8 |

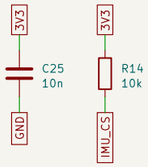
# IMU ICM-42688-P an SPI1, Barometer BMP390 und Qwiic an I2C

## IMU ICM-42688-P, SPI1





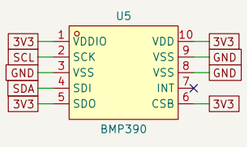


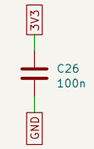


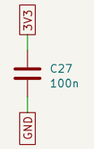


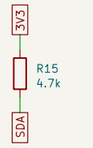


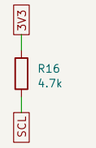
## Barometer BMP390 und I2C

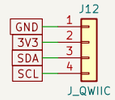



erzeugt aus circuit.py (gen_schematic.py)
IMU ICM-42688-P an SPI1, Barometer BMP390 und Qwiic an I2C
**Kartonaut**

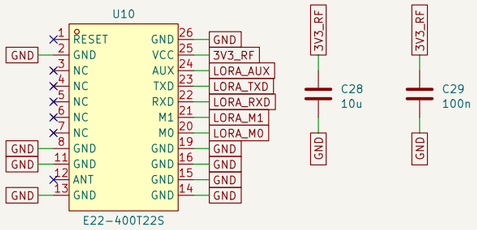
Sheet: /Sensorik/
File: sensors.kicad_sch

***Title: Mainboard rev B – Sensorik***

| Size: A3 | Date: 2026-10-21 | **Rev: B** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 4/8 |

# LoRa E22-400T22S an Serial8, GPS an Serial1

## LoRa E22-400T22S, Serial8




## GPS, Serial1

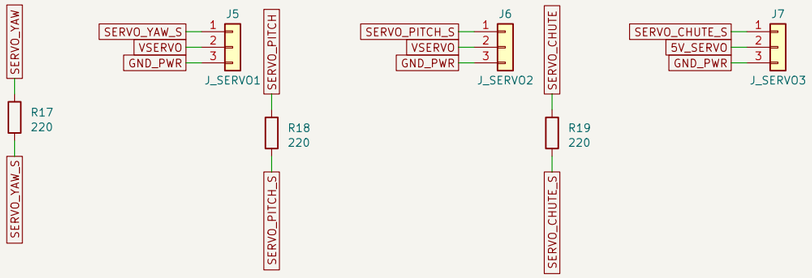



erzeugt aus circuit.py (gen_schematic.py)
LoRa E22-400T22S an Serial8, GPS an Serial1
**Kartonaut**

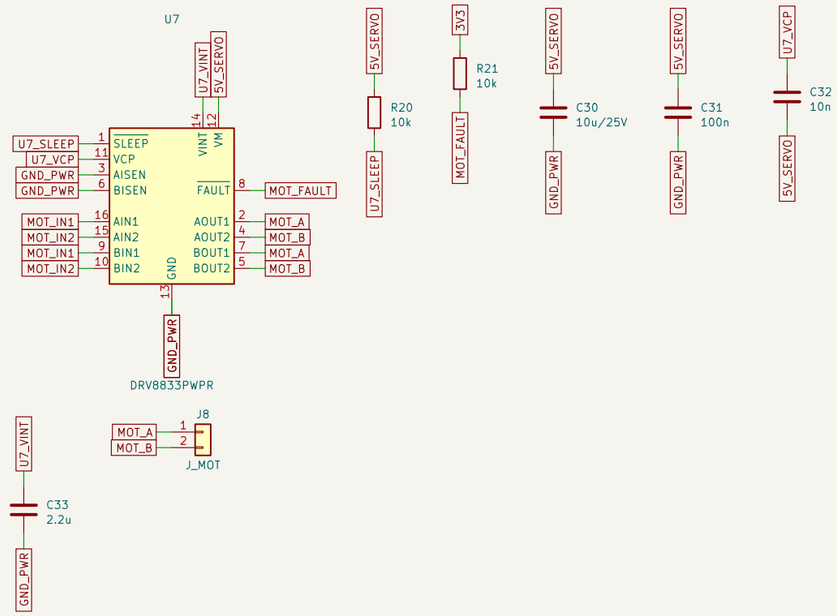
| | | |
|---|---|---|
| Sheet: /Funk/ | | |
| File: radio.kicad_sch | | |
| ***Title: Mainboard rev B – Funk*** | | |
| Size: A3 | Date: 2026-10-21 | **Rev: B** |
| KiCad E.D.A. 10.0.6 | | Id: 5/8 |

# TVC–Servos am 2S–Akku, Fallschirm–Servo, Rollmotor DRV8833

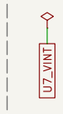
## Servos




## Rollmotor DRV8833




## Power–Flags fuer den ERC

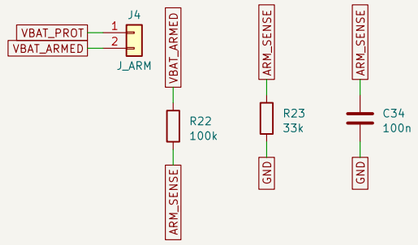



erzeugt aus circuit.py (gen_schematic.py)
TVC–Servos am 2S–Akku, Fallschirm–Servo, Rollmotor DRV8833
**Kartonaut**

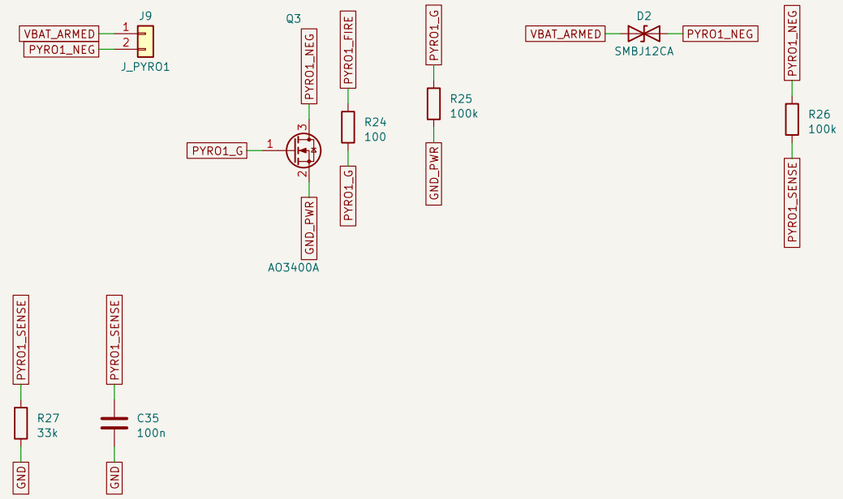
| | | |
|---|---|---|
| Sheet: /Aktoren/ | | |
| File: actuators.kicad_sch | | |
| ***Title: Mainboard rev B – Aktoren*** | | |
| Size: A3 | Date: 2026–10–21 | **Rev: B** |
| KiCad E.D.A. 10.0.6 | | Id: 6/8 |

# Zwei Pyro-Kanaele low-side, Arming, Durchgangspruefung

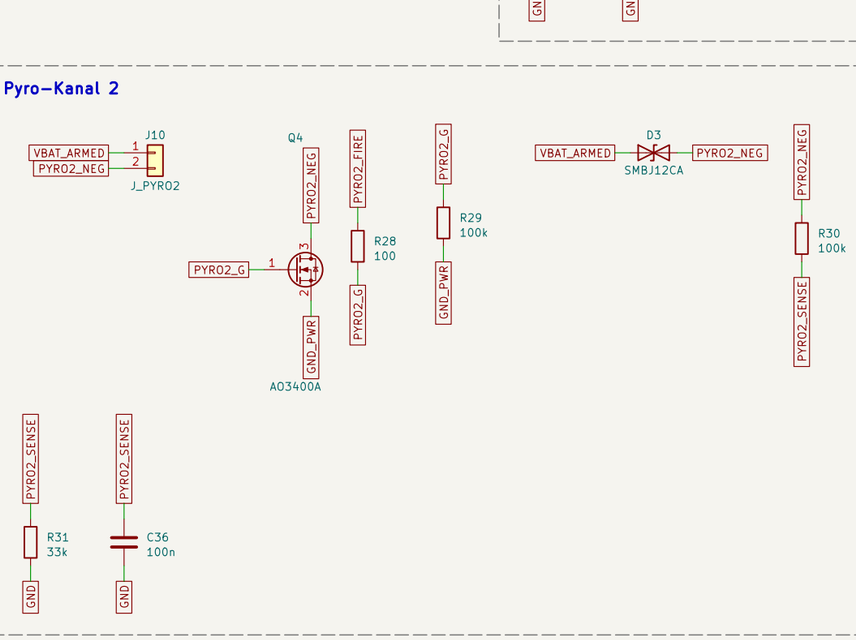
## Arming




## Pyro-Kanal 1




## Pyro-Kanal 2

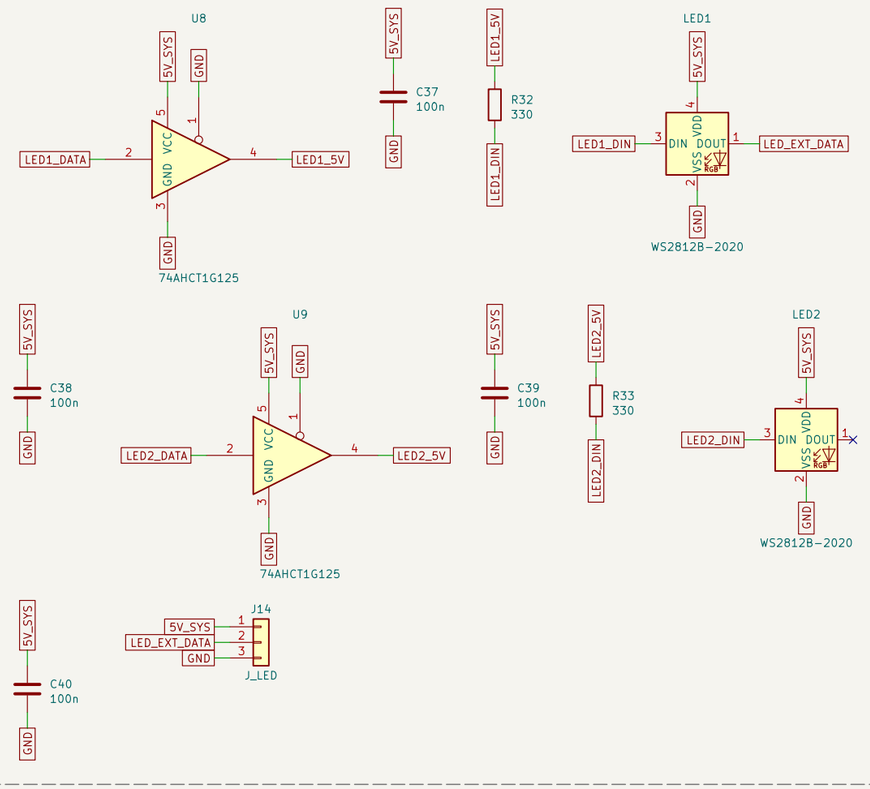



erzeugt aus circuit.py (gen_schematic.py)
Zwei Pyro-Kanaele low-side, Arming, Durchgangspruefung
**Kartonaut**

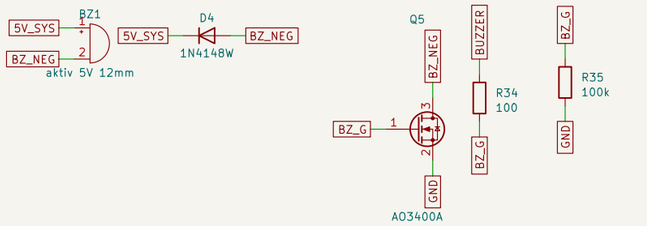
| | | |
|---|---|---|
| Sheet: /Pyro/ | | |
| File: pyro.kicad_sch | | |
| ***Title: Mainboard rev B – Pyro*** | | |
| Size: A3 | Date: 2026-10-21 | **Rev: B** |
| KiCad E.D.A. 10.0.6 | | Id: 7/8 |

# WS2812B mit Pegelwandler, Summer, Kamera–Trigger

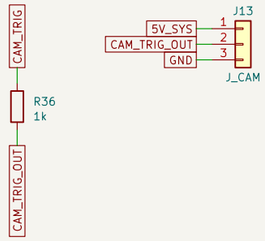
## Status–LEDs




## Summer




## Kamera




erzeugt aus circuit.py (gen_schematic.py)
WS2812B mit Pegelwandler, Summer, Kamera–Trigger
**Kartonaut**

Sheet: /Signal/
File: signal.kicad_sch

***Title: Mainboard rev B – Signal***

| | | |
|---|---|---|
| Size: A3 | Date: 2026–10–21 | **Rev: B** |
| KiCad E.D.A. 10.0.6 | | Id: 8/8 |